



Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY TECHNICAL BOOTCAMP · 40 HOURS

Agentic AI Engineering

Build production-grade AI agents, multi-agent systems, and MCP servers

5

DAYS

40

HOURS

25+

HANDS-ON LABS

6

AGENT PROJECTS

Course Overview

This 5-day technical bootcamp takes developers and architects from understanding AI agent fundamentals to building production-grade multi-agent systems using LangChain, LangGraph, CrewAI, Semantic Kernel, Microsoft Copilot Studio, and Model Context Protocol (MCP) servers. Participants build 6 real-world agent projects by the end of the program.

This is the most in-demand AI skill in 2026. Enterprises are moving beyond chatbots to autonomous AI agents that reason, plan, use tools, and orchestrate workflows. This bootcamp gives your team the hands-on ability to design, build, and deploy these systems on Azure, AWS, and open-source stacks. Designed and delivered by an AB-100 certified Agentic AI architect who has built production MCP servers.

Who Is This For?



Developers

Python, JavaScript, or .NET developers building AI features



AI Architects

Solution architects designing agentic AI systems



ML Engineers

Engineers moving from ML models to agent orchestration



Platform Engineers

DevOps and platform teams deploying AI in production

Learning Outcomes

- 1 **Build AI agents with LangChain & LangGraph** — design ReAct agents, tool-calling agents, and stateful graph-based workflows
- 2 **Orchestrate multi-agent systems with CrewAI** — create teams of specialized agents that collaborate, delegate, and reason together
- 3 **Build MCP servers from scratch** — implement Model Context Protocol servers that connect AI agents to databases, APIs, and enterprise systems
- 4 **Deploy agents on Microsoft Copilot Studio** — build enterprise-grade Copilot agents with Dataverse, Power Automate, and Azure AI integration
- 5 **Implement RAG pipelines for agents** — ground agents in enterprise knowledge using Azure AI Search, FAISS, Chroma, and document intelligence
- 6 **Design production agent architectures** — handle memory, state, human-in-the-loop, error recovery, guardrails, and monitoring

Prerequisites

Python programming (intermediate), basic understanding of REST APIs and JSON, familiarity with at least one cloud platform (Azure/AWS/GCP). Prior ML/AI experience is helpful but not required.

5-Day Curriculum

Interactive format with live coding, hands-on labs, and project-based learning. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

AI Agent Foundations — LangChain & Tool Calling

LECTURE + LIVE CODING + HANDS-ON LABS

SESSION A

Understanding AI Agents

- What are AI agents? From chatbots to autonomous systems
- Agent architectures: ReAct, Plan-and-Execute, Reflexion
- The agentic AI stack: LLMs, tools, memory, orchestration
- Azure OpenAI vs Claude vs open-source models for agents
- Lab: Set up development environment (Python, LangChain, API keys)

SESSION B

LangChain Agents & Tool Calling

- LangChain core: chains, prompts, output parsers
- Building tool-calling agents with function calling
- Custom tools: web search, database queries, API calls
- Lab: Build a research agent that searches, summarizes, and reports
- Lab: Build a SQL agent that queries databases via natural language

02

LangGraph — Stateful Multi-Step Agent Workflows

LIVE CODING + HANDS-ON LABS + ARCHITECTURE WORKSHOP

SESSION A

LangGraph Fundamentals

- Graph-based agent orchestration: nodes, edges, state
- Conditional routing and branching in agent workflows
- Persistence: checkpoints, memory, and state management
- Lab: Build a multi-step document processing agent
- Lab: Build a customer support agent with escalation logic

SESSION B

Human-in-the-Loop & Advanced Patterns

- Human-in-the-loop: approval gates, feedback loops
- Error handling, retries, and fallback strategies
- Streaming responses and real-time agent updates
- Lab: Build a loan approval agent with human review step
- Lab: Build a content moderation pipeline with multi-stage review

03

CrewAI & Multi-Agent Collaboration

LIVE CODING + HANDS-ON LABS + ARCHITECTURE DESIGN

SESSION A**CrewAI Framework Deep Dive**

- CrewAI architecture: agents, tasks, crews, and processes
- Defining agent roles, goals, and backstories
- Sequential vs hierarchical crew execution
- Lab: Build a financial research crew (analyst + writer + reviewer)
- Lab: Build an automated hiring pipeline with specialized agents

SESSION B**Semantic Kernel & Microsoft Agent Framework**

- Semantic Kernel: plugins, planners, and memory
- Microsoft Agent Framework and Azure AI Agent Service
- AutoGen: multi-agent conversation patterns
- Lab: Build a project planning agent with Semantic Kernel
- Comparing frameworks: LangChain vs CrewAI vs Semantic Kernel

04

MCP Servers & Enterprise Integration

LIVE CODING + HANDS-ON LABS + PRODUCTION PATTERNS

SESSION A**Model Context Protocol (MCP) Servers**

- What is MCP? Architecture, tools, resources, and prompts
- Building MCP servers with Python and FastAPI
- Connecting MCP servers to databases (SQL, Cosmos DB)
- Lab: Build an Azure SQL MCP server from scratch
- Lab: Build a document retrieval MCP server with AI Search

SESSION B**Copilot Studio & Enterprise Deployment**

- Connecting MCP servers to Microsoft Copilot Studio
- Building enterprise agents with Dataverse integration
- Power Automate flows triggered by AI agents
- Lab: Deploy MCP server to Azure and connect to Copilot Studio
- Lab: Build insurance claims processing agent (end-to-end)

05

RAG, Production Architecture & Capstone

CAPSTONE PROJECT + ARCHITECTURE REVIEW + PRESENTATIONS

SESSION A**RAG Pipelines for Agents**

- Retrieval-Augmented Generation: chunking, embedding, indexing
- Azure AI Search, FAISS, Chroma, Pinecone integration
- Hybrid search: keyword + semantic + reranking
- Lab: Build a RAG-powered knowledge agent for enterprise docs
- Agent guardrails: content safety, prompt injection defense

SESSION B**Capstone: Production Agent System**

- Team project: Design and build a complete agentic AI system
- Architecture review: scalability, monitoring, cost optimization
- CI/CD for agents: testing, versioning, deployment pipelines
- Team presentations and expert feedback
- Certification paths and next steps

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools used hands-on with live coding and lab exercises

LangChain

LangGraph

CrewAI

Semantic Kernel

AutoGen

MS Agent Framework

MCP Servers

Microsoft Copilot Studio

Azure OpenAI

Azure AI Search

FAISS / Chroma

FastAPI

Power Automate

Dataverse

Python

Docker

What You Take Home

**6 Agent Projects**
Complete source code for all lab projects — research agent, SQL agent, loan agent, claims agent, RAG agent, MCP server

**Architecture Templates**
Production-ready agent architecture blueprints for BFSI, Government, Healthcare, and IT sectors

**MCP Server Starter Kit**
Open-source MCP server template with Azure SQL, FastAPI, and Copilot Studio connection guide

**Agent Design Canvas**
Framework for designing multi-agent systems: roles, tools, memory, guardrails, and deployment checklist

Certification of Completion

**Gennoor Tech — Certificate of Completion**
Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "Agentic AI Engineering". Digitally signed by the Founder & Director. Validates proficiency in AI agent development, multi-agent orchestration, MCP servers, and production deployment patterns.

Recommended Next Certifications

Microsoft AB-100
Agentic AI Business Solutions Architect

Microsoft AI-103
Azure AI App & Agent Developer Associate

AI-3026
Develop AI Agents on Azure

AZ-2005
AI Agents & Semantic Kernel

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited

AB-100 Certified Agentic AI Business Solutions Architect, Senior AI Consultant, and Microsoft Certified Trainer (MCT) with 14+ years of experience. Built and open-sourced the Azure SQL MCP Server connecting Copilot Studio to enterprise databases. Delivered the 40-hour Agentic AI in Power Platform program for EY BFSI developers.

16+ active Microsoft certifications including AI-103, AZ-400, AI-300, PL-400, and GitHub certifications. Trained teams at Microsoft, IBM, Boeing, Saudi Aramco, Bank of Tanzania, and MCIT Saudi Arabia.
M.E. Mechanical Engineering, Mumbai University.

Delivery Options

 **On-Site** — customized at your office |  **Virtual Live** — interactive via Teams/Zoom |  **International** — GCC, Africa, APAC, UK



Ready to Build Autonomous AI Agents?

Contact us to discuss a customized bootcamp
for your organization or register for the next open batch.

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